CHAPTER 16: THE LEARNING CONTINUES

Designing good systems requires years of accumulation of knowledge. One shortcut is to dive into real-world system architectures. Below is a collection of helpful reading materials. We highly recommend you pay attention to both the shared principles and the underlying technologies. Researching each technology and understanding what problems it solves is a great way to strengthen your knowledge base and refine the design process.

Real-world systems

The following materials can help you understand general design ideas of real system architectures behind different companies.

Facebook Timeline: Brought To You By The Power Of Denormalization: https://goo.gl/FCNrbm

Scale at Facebook: https://goo.gl/NGTdCs

Building Timeline: Scaling up to hold your life story: https://goo.gl/8p5wDV

Erlang at Facebook (Facebook chat): https://goo.gl/zSLHrj

Facebook Chat: https://goo.gl/qzSiWC

Finding a needle in Haystack: Facebook's photo storage: https://goo.gl/edj4FL

Serving Facebook Multifeed: Efficiency, performance gains through redesign: https://goo.gl/adFVMQ

Scaling Memcache at Facebook: https://goo.gl/rZiAhX

TAO: Facebook's Distributed Data Store for the Social Graph: https://goo.gl/Tk1DyH

Amazon Architecture: https://goo.gl/k4feoW

Dynamo: Amazon's Highly Available Key-value Store: https://goo.gl/C7zxDL

A 360 Degree View Of The Entire Netflix Stack: https://goo.gl/rYSDTz

It's All A/Bout Testing: The Netflix Experimentation Platform: https://goo.gl/agbA4K

Netflix Recommendations: Beyond the 5 stars (Part 1): https://goo.gl/A4FkYi

Netflix Recommendations: Beyond the 5 stars (Part 2): https://goo.gl/XNPMXm

Google Architecture: https://goo.gl/dvkDiY

The Google File System (Google Docs): https://goo.gl/xj5n9R

Differential Synchronization (Google Docs): https://goo.gl/9zqG7x

YouTube Architecture: https://goo.gl/mCPRUF

Seattle Conference on Scalability: YouTube Scalability: https://goo.gl/dH3zYq

Bigtable: A Distributed Storage System for Structured Data: https://goo.gl/6NaZca

Instagram Architecture: 14 Million Users, Terabytes Of Photos, 100s Of Instances, Dozens Of Technologies: https://goo.gl/s1VcW5

The Architecture Twitter Uses To Deal With 150M Active Users: https://goo.gl/EwvfRd

Scaling Twitter: Making Twitter 10000 Percent Faster: https://goo.gl/nYGC1k

Announcing Snowflake (Snowflake is a network service for generating unique ID numbers at high scale with some simple guarantees): https://goo.gl/GzVWYm

Timelines at Scale: https://goo.gl/8KbqTy

How Uber Scales Their Real-Time Market Platform: https://goo.gl/kGZuVy

Scaling Pinterest: https://goo.gl/KtmjW3

Pinterest Architecture Update: https://goo.gl/w6rRsf

A Brief History of Scaling LinkedIn: https://goo.gl/8A1Pi8

Flickr Architecture: https://goo.gl/dWtgYa

How We've Scaled Dropbox: https://goo.gl/NjBDtC

The WhatsApp Architecture Facebook Bought For $19 Billion: https://bit.ly/2AHJnFn

Company engineering blogs

If you are going to interview with a company, it is a great idea to read their engineering blogs and get familiar with technologies and systems adopted and implemented there. Besides, engineering blogs provide invaluable insights about certain fields. Reading them regularly could help us become better engineers.

Here is a list of engineering blogs of well-known large companies and startups.

Airbnb: https://medium.com/airbnb-engineering
Amazon: https://developer.amazon.com/blogs
Asana: https://blog.asana.com/category/eng
Atlassian: https://developer.atlassian.com/blog
Bittorrent: http://engineering.bittorrent.com
Cloudera: https://blog.cloudera.com
Docker: https://blog.docker.com
Dropbox: https://blogs.dropbox.com/tech
eBay: http://www.ebaytechblog.com
Facebook: https://code.facebook.com/posts
GitHub: https://githubengineering.com
Google: https://developers.googleblog.com
Groupon: https://engineering.groupon.com
Highscalability: http://highscalability.com
Instacart: https://tech.instacart.com
Instagram: https://engineering.instagram.com
Linkedin: https://engineering.linkedin.com/blog
Mixpanel: https://mixpanel.com/blog
Netflix: https://medium.com/netflix-techblog
Nextdoor: https://engblog.nextdoor.com
PayPal: https://www.paypal-engineering.com
Pinterest: https://engineering.pinterest.com
Quora: https://engineering.quora.com
Reddit: https://redditblog.com
Salesforce: https://developer.salesforce.com/blogs/engineering
Shopify: https://engineering.shopify.com
Slack: https://slack.engineering
Soundcloud: https://developers.soundcloud.com/blog
Spotify: https://labs.spotify.com
Stripe: https://stripe.com/blog/engineering
System design primer: https://github.com/donnemartin/system-design-primer
Twitter: https://blog.twitter.com/engineering/en_us.html
Thumbtack: https://www.thumbtack.com/engineering
Uber: http://eng.uber.com

Yahoo: https://yahooeng.tumblr.com
Yelp: https://engineeringblog.yelp.com
Zoom: https://medium.com/zoom-developer-blog

AFTERWORD

Congratulations! You are at the end of this interview guide. You have accumulated skills and knowledge to design systems. Not everyone has the discipline to learn what you have learned. Take a moment and pat yourself on the back. Your hard work will be paid off.

Landing a dream job is a long journey and requires lots of time and effort. Practice makes perfect. Best luck!

Thank you for buying and reading this book. Without readers like you, our work would not exist. We hope you have enjoyed the book!

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Please subscribe to our email list if you want to be notified when new chapters are available: https://bit.ly/3dtIcsE

If you have comments or questions about this book, feel free to send us an email at systemdesigninsider@gmail.com. Besides, if you notice any errors, please let us know so we can make corrections in the next edition. Thank you!